FEDERAL STATE BUDGETARY EDUCATIONAL INSTITUTION OF HIGHER EDUCATION «UFA UNIVERSITY OF SCIENCE AND TECHNOLOGY»

INSTITUTE OF COMPUTER SCIENCE, MATHEMATICS AND ROBOTICS

MATHEMATICAL AND COMPUTER MODELING DEPARTMENT

Technology (project-technology) practice
Report

Completed by

the 3d year full-time student

of the field of study (specialty):

Applied Mathematics and Computer Science

Focus (profile):

Programming and Data Analysis

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of Mathematics and computer modeling

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Top Toys is planning a new radio and TV advertising campaign. A radio commercial costs $300 and a TV ad costs $2000. A total budget of $20,000 is allocated to the campaign. However, to ensure that each medium will have at least one radio commercial and one TV ad, the most that can be allocated to either medium cannot exceed 80% of the total budget. It is estimated that the first radio commercial will reach 5000 people, with each additional commercial reaching only 2000 new ones. For TV, the first ad will reach 4500 people, and each additional ad an additional 3000. How should the budgeted amount be allocated between radio and TV?---------- :

1. Mathematical Model Formulation

Decision Variables

- Let $x_1 =$ the number of radio commercials purchased.
- Let $x_2 =$ the number of TV ads purchased.

Objective Function

We want to maximize the total audience reached. [1]

- **Radio reach**: The first ad reaches $5,000$ people. The remaining $(x_1 - 1)$ ads reach $2,000$ people each.
  $\text{Radio Reach}=5000+2000(x_{1}-1)=2000x_{1}+3000$
- **TV reach**: The first ad reaches $4,500$ people. The remaining $(x_2 - 1)$ ads reach $3,000$ people each.
  $\text{TV Reach}=4500+3000(x_{2}-1)=3000x_{2}+1500$ [1, 2, 3]

Combining these two terms yields the objective function:
$\text{Maximize }Z=2000x_{1}+3000x_{2}+4500$

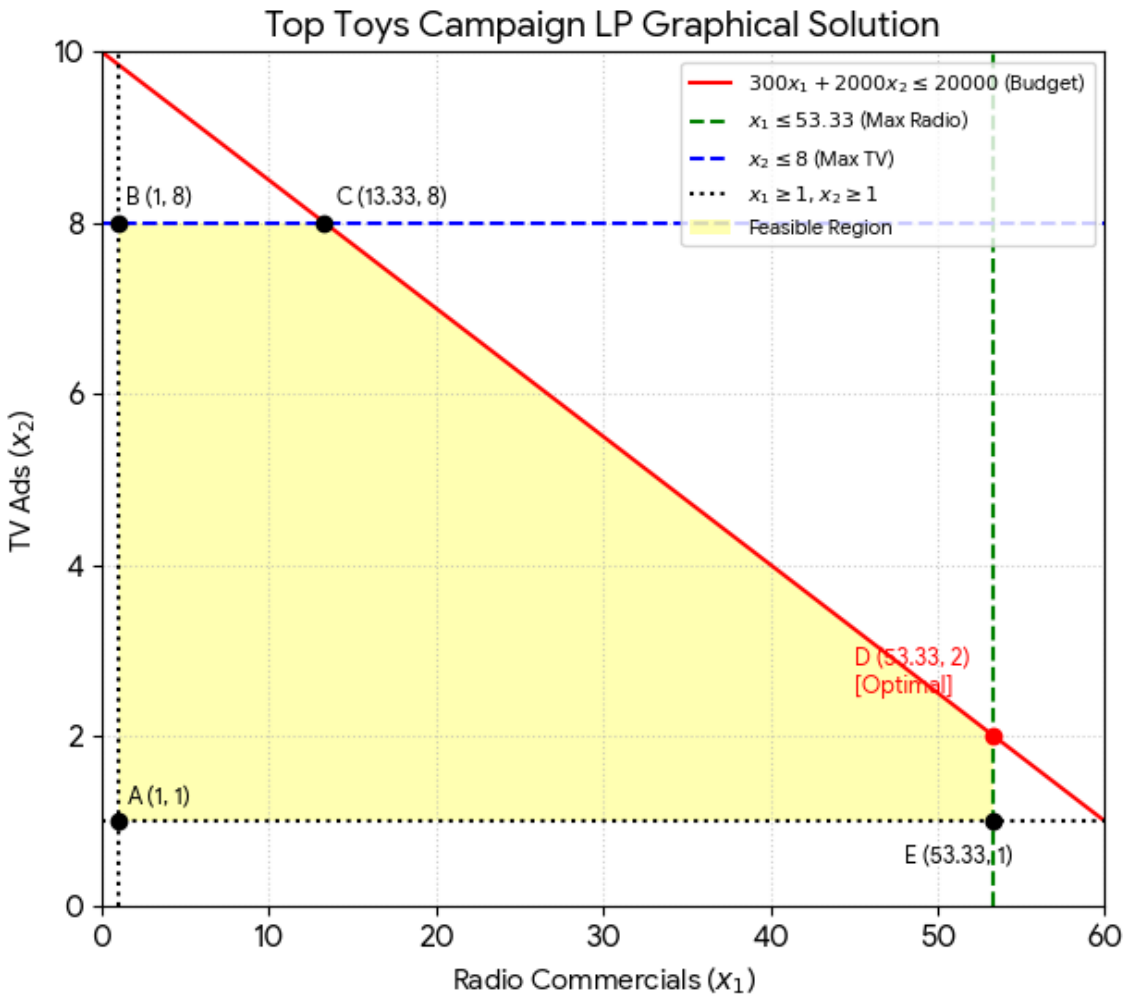
Constraints

1. **Total Budget Limit**: The overall cost cannot exceed $\$20,000$.
   $300x_{1}+2000x_{2}\le 20000$
2. **Maximum Radio Budget**: No more than 80% of the total budget ($\$16,000$) can go to radio.
   $300x_{1}\le 16000\implies x_{1}\le 53.33$
3. **Maximum TV Budget**: No more than 80% of the total budget ($\$16,000$) can go to TV.
   $2000x_{2}\le 16000\implies x_{2}\le 8$
4. **Minimum Quantities**: To guarantee at least one ad per medium:
   $x_{1}\ge 1$
   $x_{2}\ge 1$ [1, 2]

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2. Graphical Solution

The shaded region represents the feasible space dictated by the upper and lower boundary lines. The optimal solution must occur at one of the corner vertices. [1]



3. Corner Point Analysis & Optimal Plan

Evaluating the exact audience reach (\(Z\)) at each extreme corner point of our feasible region:

- **Point A \((1, 1)\)**: \(Z = 2000(1) + 3000(1) + 4500 =\) **\(9,500\) people** (Cost: \(\$2,300\))
- **Point B \((1, 8)\)**: \(Z = 2000(1) + 3000(8) + 4500 =\) **\(30,500\) people** (Cost: \(\$16,300\))
- **Point C \((13.33, 8)\)**: \(Z = 2000(13.33) + 3000(8) + 4500 =\) **\(55,167\) people** (Cost: \(\$20,000\))

- **Point D $((53.33, 2))$**: $Z = 2000(53.33) + 3000(2) + 4500 =$ **$(117,167)$ people** (Cost: $\$20,000$)
- **Point E $((53.33, 1))$**: $Z = 2000(53.33) + 3000(1) + 4500 =$ **$(114,167)$ people** (Cost: $\$18,000$)

John must work at least 20 hours a week to supplement his income while attending school. He has the opportunity to work in two retail stores. In store 1, he can work between 4.5 and 12 hours a week, and in store 2, he is allowed between 5.5 and 10 hours. Both stores pay the same hourly wage. In deciding how many hours to work in each store, John wants to base his decision on work stress. Based on interviews with present employees :----------

1. Mathematical Model Formulation

Decision Variables

- Let $x_1 =$ number of hours worked per week in Store 1.
- Let $x_2 =$

- **Objective Function**
- John wants to minimize his total weekly work stress: $\text{Minimize }Z=8x_{1}+6x_{2}$

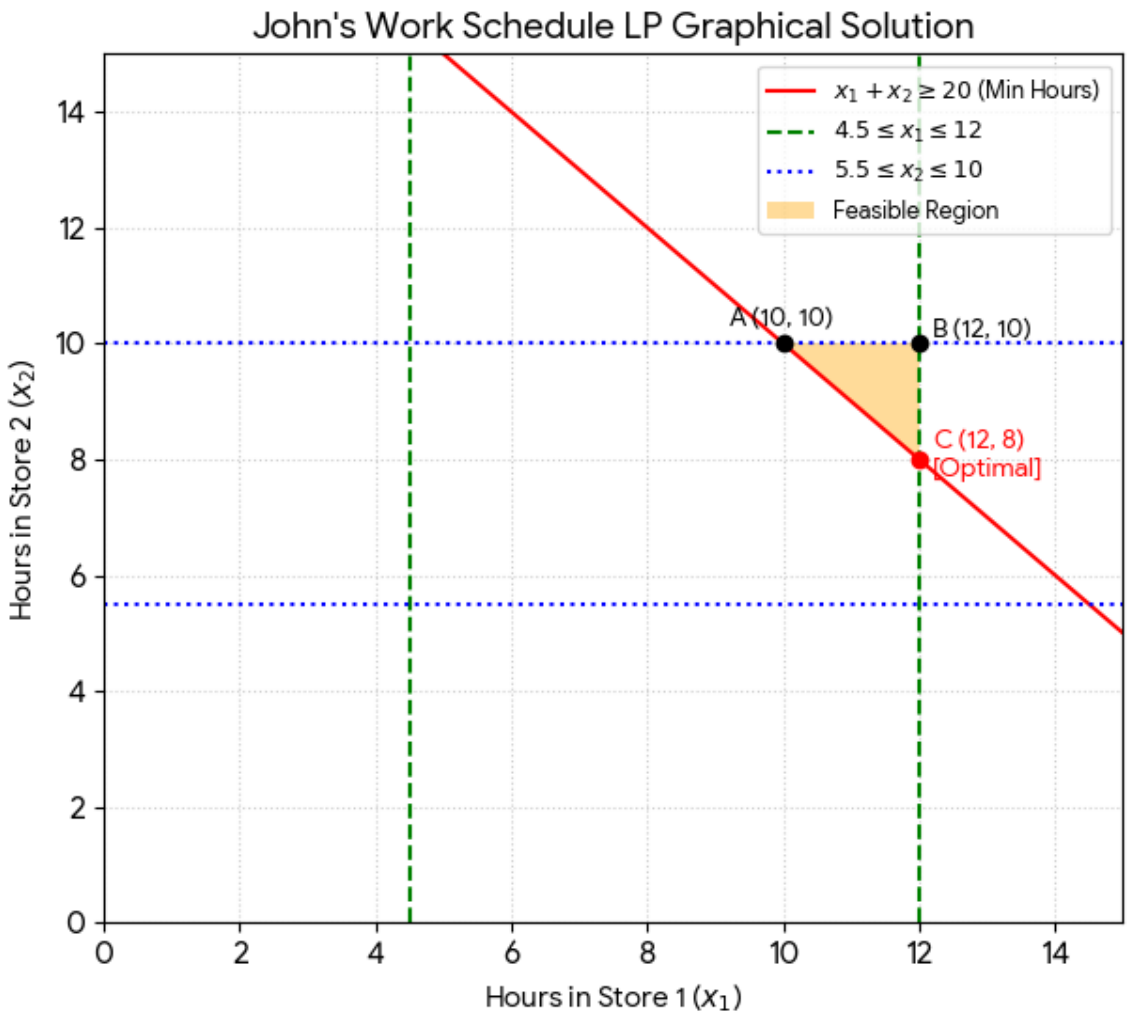
Constraints

1. **Minimum Total Work Hours**: He must work at least 20 hours a week. $x_{1}+x_{2}\ge 20$
2. **Store 1 Hours Allowed**: He can work between 4.5 and 12 hours. $4.5\le x_{1}\le 12$
3. **Store 2 Hours Allowed**: He can work between 5.5 and 10 hours. $5.5\le x_{2}\le 10$

\(5.5\le x_{2}\le 10\)

2. Graphical Solution

The shaded polygon highlights the area where all boundary lines intersect. To minimize stress, our objective line slides toward the lower left limits of this zone.



3. Corner Point Analysis & Optimal Plan

Let's test the extreme corner vertices of the feasible region inside our objective function: [1]

- **Point A \((10, 10)\)**: \(Z = 8(10) + 6(10) =\) **\(140\) stress points** (Total hours: \(20\))
- **Point B \((12, 10)\)**: \(Z = 8(12) + 6(10) =\) **\(156\) stress points** (Total hours: \(22\))
- **Point C \((12, 8)\)**: \(Z = 8(12) + 6(8) =\) **\(144\) stress points** (Total hours: \(20\))

Wait, let's verify if **Point A** is the minimum!

- \(Z\) at Point A \((10,10) = 140\)
- \(Z\) at Point C \((12,8) = 144\)

Conclusion & Schedule Plan

The minimum objective stress value is **140**, occurring at **Point A**. [1, 2]

- **Store 1 Allocation**: John should work **10 hours** per week.
- **Store 2 Allocation**: John should work **10 hours** per week.
- **Total Work Time**: **20 hours** per week (exactly satisfying his school-supplement requirement).
- **Total Weekly Stress Score**: **140**. [1, 2]

Determine the feasible space for each of the following independent constraints, given that x1, x2 Ú 0---------- :

*(a) : -3x_1 + x_2 <= 6 :

. Step-by-Step Mathematical Analysis

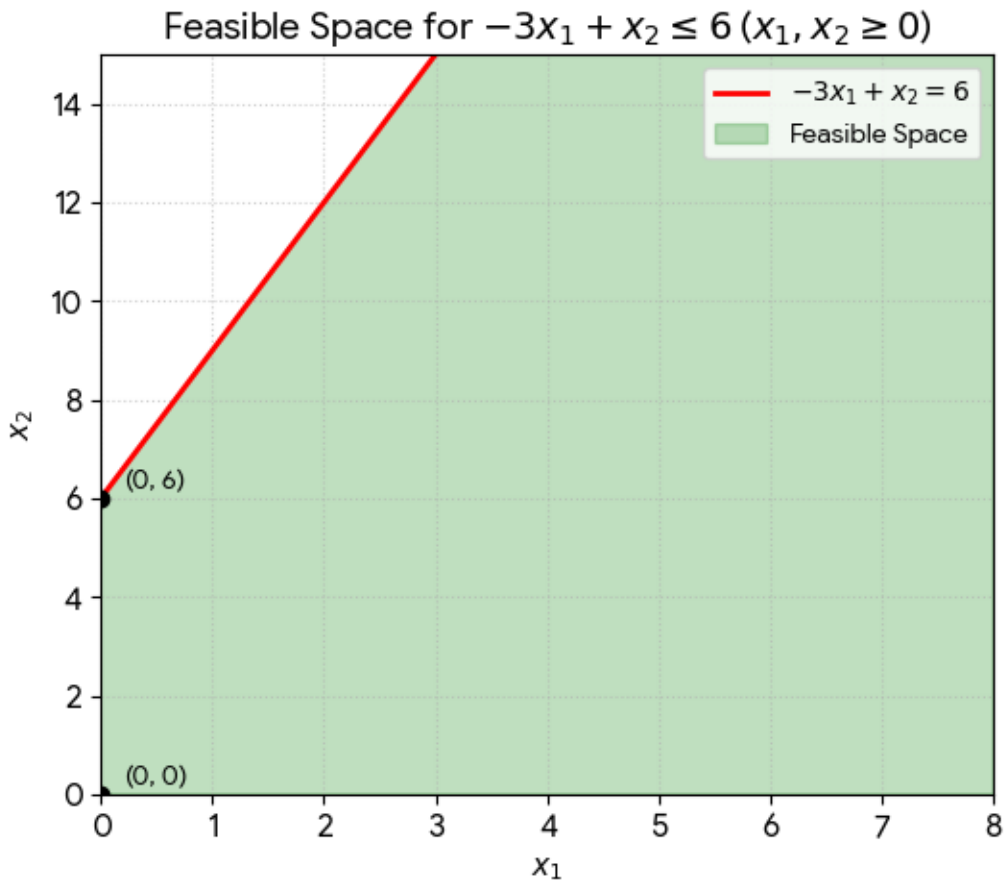
- **Step 1: Convert the inequality to a boundary equation.**
  \(-3x_{1}+x_{2}=6\)

- **Step 2: Find the intercepts.**
    - Set $x_1 = 0$:
      $-3(0) + x_2 = 6 \implies \mathbf{x_2 = 6}$. This gives the vertical axis intercept at **$(0, 6)$**.
    - Set $x_2 = 0$:
      $-3x_1 + 0 = 6 \implies \mathbf{x_1 = -2}$. This gives a horizontal intercept at **$(-2, 0)$**. *(Since we are restricted to $x_1 \geq 0$, this point is outside our plotting region but helps guide the slope of the line).*
- **Step 3: Test a point to determine the feasible direction.**
  Test the origin $(0,0)$:
  $-3(0)+(0)\le 6\implies 0\le 6\quad \text{[True]}$
  Since the statement is true, the region containing the origin $(0,0)$ is the valid space.

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2. Graphical Drawing

The feasible space is bounded by the axes $x_1 \geq 0$, $x_2 \geq 0$ and the upward-sloping boundary line. It creates an **unbounded feasible region** stretching infinitely to the right.



*(b) : x_1 - 2x_2 >= 5 :

1. Step-by-Step Mathematical Analysis

Given that **\(x_1, x_2 \geq 0\)**:

- **Step 1: Convert the inequality to a boundary equation.**
  \(x_{1}-2x_{2}=5\)
- **Step 2: Find the intercepts.**
  - Set \(x_2 = 0\):
    \(x_{1}-2(0)=5\implies \mathbf{x}_{\mathbf{1}}\mathbf{=5}\)
    This gives the horizontal axis intercept at **\((5, 0)\)**.

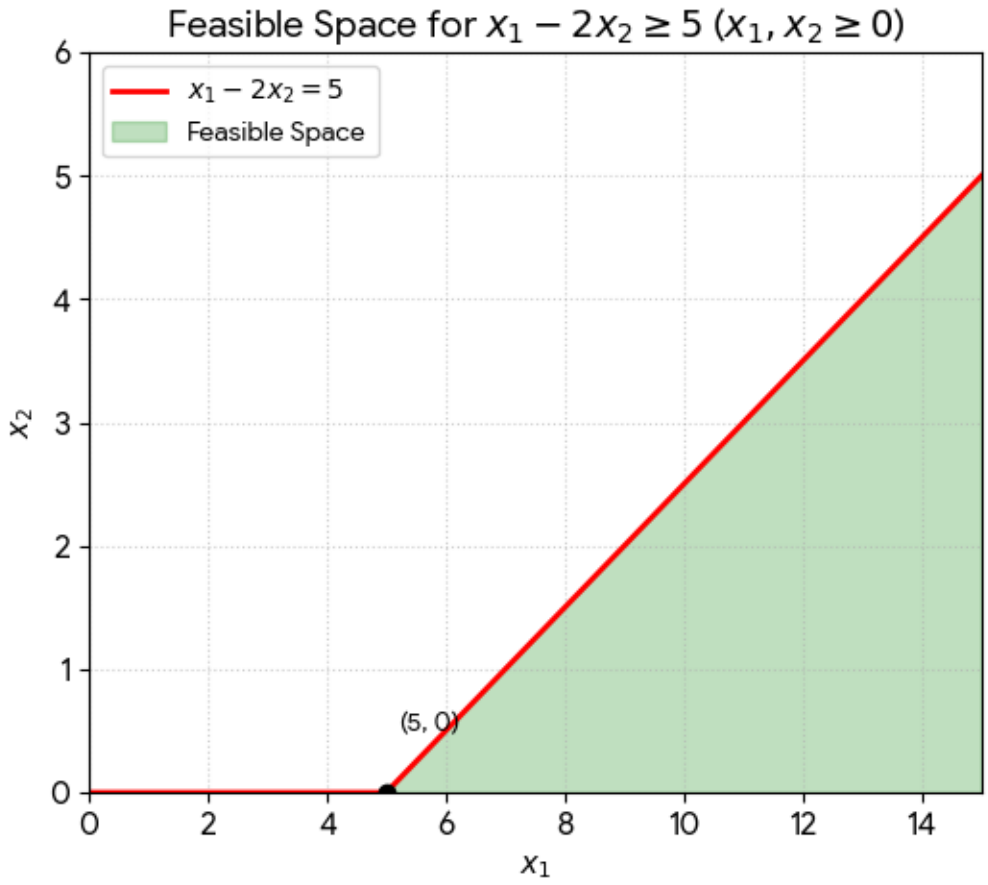
- Set \(x_1 = 0\):
  \(0-2x_{2}=5\implies \mathbf{x}_{\mathbf{2}}\mathbf{=-2.5}\)
  This intercept lies outside the first quadrant, but it tells us the line slopes upward as it goes to the right.

- **Step 3: Test a point to determine the feasible direction.**
  Test the origin \((0,0)\):
  \(0-2(0)\ge 5\implies 0\ge 5\quad \text{[False]}\)
  Since the statement is false, the feasible space lies **away** from the origin (to the right of the line).

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2. Graphical Drawing

The feasible space is bounded on the left by the line \(x_1 - 2x_2 = 5\) and on the bottom by the horizontal axis (\(x_2 \geq 0\)). This creates an **unbounded region** expanding infinitely to the right.



*(c) : 2x_1 - 3x_2 <= 12 :

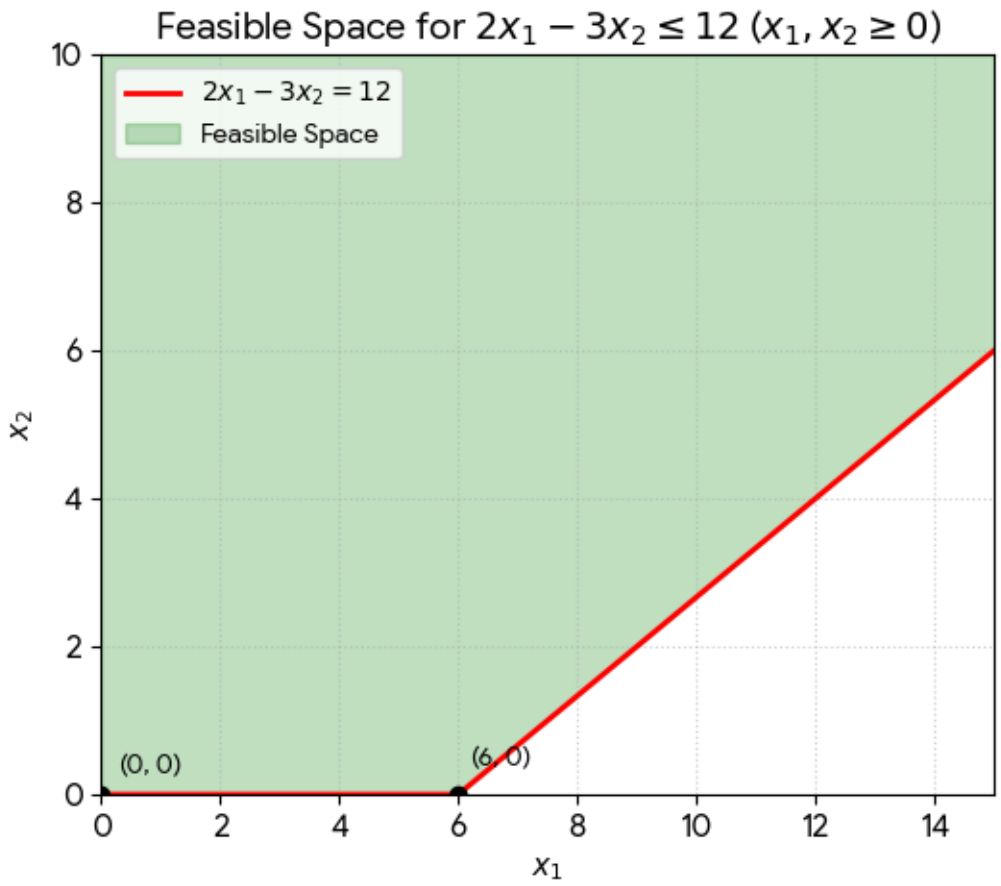
1. Step-by-Step Mathematical Analysis

- **Step 1: Convert the inequality to a boundary equation.**
  \(2x_{1}-3x_{2}=12\)
- **Step 2: Find the axis intercepts.**
  - Set \(x_2 = 0\):
    \(2x_{1}-3(0)=12\implies 2x_{1}=12\implies \mathbf{x}_{\mathbf{1}}\mathbf{=6}\)
    This gives the horizontal axis intercept at **\((6, 0)\)**.
  - Set \(x_1 = 0\):
    \(2(0)-3x_{2}=12\implies -3x_{2}=12\implies \mathbf{x}_{\mathbf{2}}\mathbf{=-4}\)
    This intercept lies below the horizontal axis outside the first quadrant, showing that the line rises steadily as it moves right.

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2. Graphical Drawing

The feasible area is bounded on the bottom right by the boundary line. It stays open and unbounded traveling straight upward and toward the left within the positive quadrant.



. Feasible Space Summary

- **Lower Right Boundary**: Bounded by the line \(2x_1 - 3x_2 = 12\) starting out from \(x_1 = 6\).
- **Left Boundary**: Bounded by the vertical axis (\(x_1 \geq 0\)).
- **Bottom Left Boundary**: Bounded by the horizontal axis (\(x_2 \geq 0\)) from \(x_1 = 0\) to \(x_1 = 6\).
- **Characteristics**: This space is **unbounded**. You can increase \(x_{2}\) infinitely upward without ever crossing the constraint barrier.

*(d) : x_1 - x_2 <= 0 :

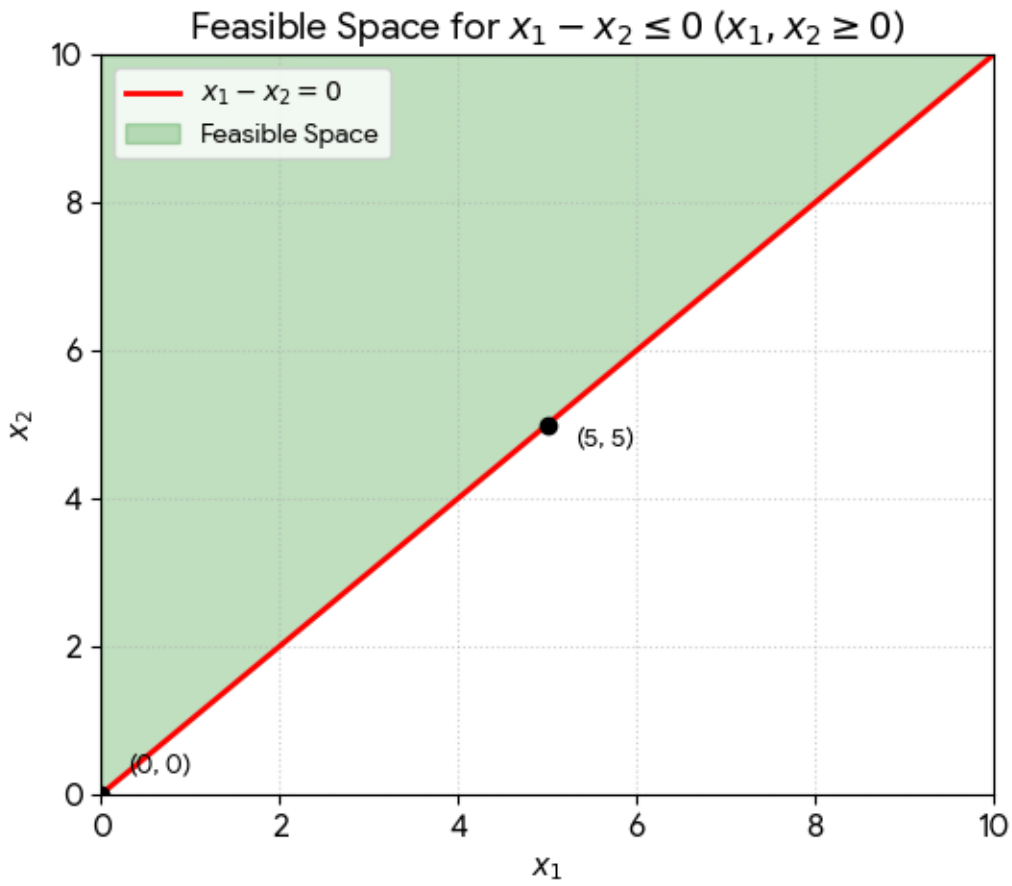
1. Step-by-Step Mathematical Analysis

- **Step 1: Convert the inequality to a boundary equation.**
  $x_{1}-x_{2}=0\implies x_{2}=x_{1}$
- **Step 2: Find the axis intercepts.**
  Since the equation is $x_2 = x_1$, setting either variable to 0 results in the other being 0. The boundary line passes directly through the origin **$(0, 0)$**. To plot it, we can pick any other equal coordinate point, such as **$(5, 5)$** or **$(10, 10)$**. This forms a perfect 45-degree diagonal line splitting the quadrant.
- **Step 3: Test a point to find the valid side.**
  Because the line passes through the origin $(0,0)$, we cannot use it as a test point. Instead, let's test a point clearly on one side of the line, like **$(0, 5)$** on the upper-left side:
  $0-5\le 0\implies -5\le 0\quad \text{[True]}$
  Since this statement is true, the region **above and to the left** of the diagonal line forms our feasible space. Rearranging the original inequality to **$x_2 \geq x_1$** confirms this visually (the vertical height x_{2} must always be greater than or equal to the horizontal distance x_{1}).

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2. Graphical Drawing

The feasible area is bounded on the bottom-right by the 45-degree boundary line and on the left by the vertical axis ($x_1 \geq 0$). It extends infinitely upward and to the right.



Feasible Space Summary

- **Bottom-Right Boundary**: Bounded by the 45-degree diagonal line \(x_2 = x_1\).
- **Left Boundary**: Bounded by the vertical axis (\(x_1 = 0\)).
- **Characteristics**: This space is **unbounded**. You can increase both variables simultaneously or increase \(x_{2}\) independently toward infinity without ever leaving the feasible region.

*(e) : -x_1 + x_2 >= 0 :

1. Step-by-Step Mathematical Analysis

- **Step 1: Convert the inequality to a boundary equation.**
  \(-x_{1}+x_{2}=0\implies x_{2}=x_{1}\) [1, 2]

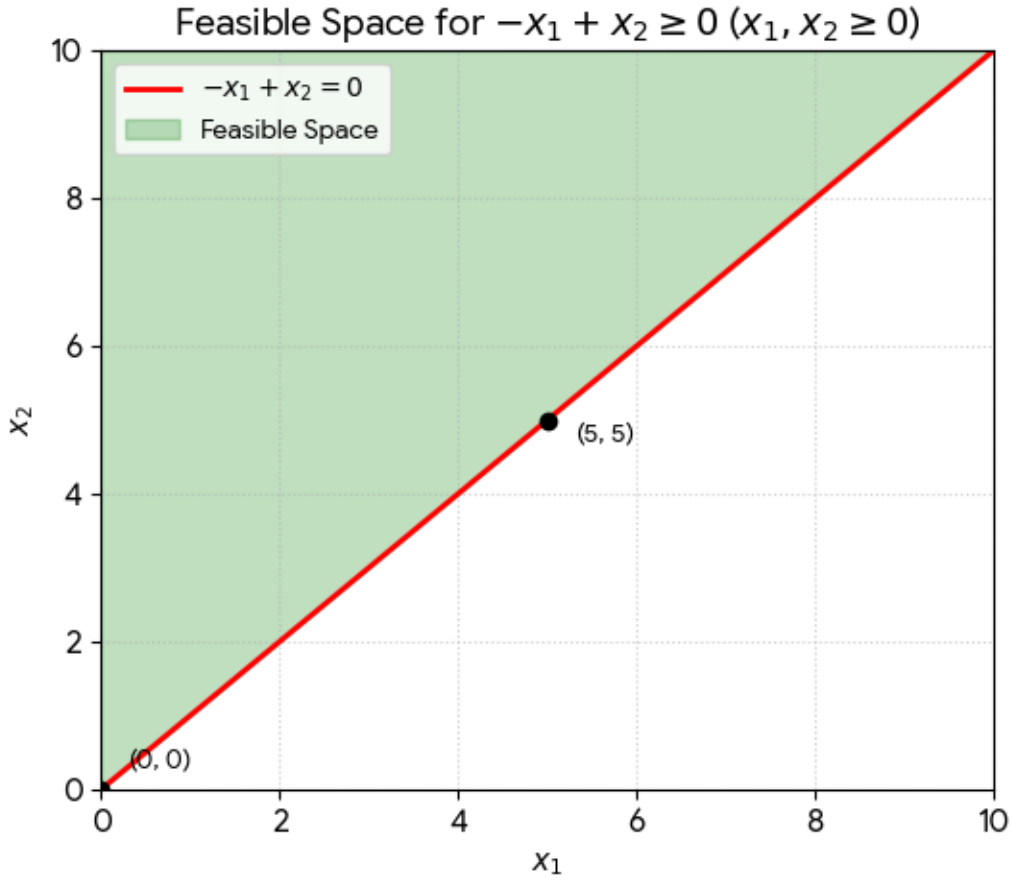
- **Step 2: Find the axis intercepts.**
  Just like constraint (d), this line passes directly through the origin **\((0, 0)\)**. It is a 45-degree diagonal line running through the first quadrant where every coordinate value satisfies \(x_1 = x_2\) (such as \((5,5)\), \((10,10)\), etc.). [1]

- **Step 3: Test a point to find the valid side.**
  Since the line runs directly through the origin, we pick an off-line point to test. Let's choose **\((0, 5)\)** (located on the upper-left vertical axis):
  \(-(0)+5\ge 0\implies 5\ge 0\quad \text{[True]}\)
  Because this statement is true, the region **above and to the left** of the diagonal line forms our feasible space.*Note: Rearranging the original inequality by adding \(x_{1}\) to both sides gives **\(x_2 \geq x_1\)**. This proves that constraint (e) is mathematically and geometrically identical to constraint (d).* [1, 2, 3]

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2. Graphical Drawing

The feasible area is bounded on the bottom-right by the 45-degree boundary line and on the left by the vertical axis (\(x_1 \geq 0\)). It extends infinitely upward and to the right. [1, 2]



Feasible Space Summary

- **Bottom-Right Boundary**: Bounded by the 45-degree diagonal line \(x_2 = x_1\).
- **Left Boundary**: Bounded by the vertical axis (\(x_1 = 0\)).
- **Characteristics**: The space is **unbounded** and identical to problem (d).